

CONDITION (C4) AND THE NESTED/PARTITION DICHOTOMY: WHY RAMANUJAN BOUNDS CANNOT DECIDE NON-AMENABILITY OF THE HURWITZ GROUPOID

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ABSTRACT. Condition (C4) of Paper X of the series asks whether the measured groupoid of the Hurwitz Bost–Connes system is non-amenable, equivalently whether the associated factor escapes the injectivity that collapsed every earlier model (Paper IX). We examine the natural route — Lubotzky–Phillips–Sarnak generators [4], the Bruhat–Tits tree, Deligne’s bound [2], a spectral gap for a Hecke–Markov operator in the sense of Kesten [3] — and show that it is closed, for reasons that are structural rather than technical.

The obstruction is a dichotomy. A $(p+1)$ -regular tree adjacency operator requires the $p+1$ branches at a vertex to *partition*; but the Ore and LCM conditions of Paper X force the range projections to be *nested*, $e_j e_k = e_{a_j \vee a_k} = e_p \neq 0$ — the central element p lies in every maximal right ideal of norm p — with $\varphi(e_j e_k) = p^{-2\beta} > 0$ and $\sum_k \varphi(e_{a_k}) = (p+1)p^{-\beta} \neq 1$. The very property that keeps the high-temperature region alive — absence of a Cuntz relation — is the property that destroys the tree.

Forcing the partition does not help, and the reason is sharper than a numerical mismatch: in any unital algebra generated by isometries v_a with $v_a v_b = v_{ab}$, orthogonality of two range projections of norm p forces $e_p = 0$ and hence $1 = v_p^* v_p = 0$. *The boundary quotient of the Hurwitz system is the zero algebra.* Even a partition in mean, $\sum_k \varphi(e_{a_k}) = 1$, would force $\beta = \log(p+1)/\log p$, a value that varies with p (1.2619, 1.1133, 1.0686, ... for $p = 3, 5, 7$). Tree structure and thermodynamics are mutually exclusive here.

Two further points are recorded. The proposed operator $\mathcal{M}_p = \frac{1}{p+1} \sum_k v_{a_k}^* (\cdot) v_{a_k}$ is unital completely positive but does not preserve φ : one has $\varphi \circ \mathcal{M}_p \leq p^\beta \varphi$ with the naive bound $\|\mathcal{M}_p\|_{L^2(\varphi)} \leq p^{\beta/2} > 1$, so no contraction is available, let alone a gap below $2\sqrt{p}/(p+1)$. And the statement of the spectral gap is not well posed: for a non-singular action the Koopman representation satisfies $\pi(a)1 = N(a)^{\beta/2} 1_{a\Omega} \neq 1$, so C1 is not invariant and $L_0^2 = L^2 \ominus \text{C1}$ is not a subrepresentation.

We close with the reformulation we believe is correct: by Takesaki duality the factor is injective if and only if its continuous core is, so (C4) is equivalent to non-amenable of the *Maharam extension*, which is measure preserving. That does not settle it — non-amenable groups do have amenable infinite-measure-preserving actions — but it places the question in a measure-preserving setting, where the obstruction identified here does not apply. Condition (C4) has since been settled, in the negative, in Paper XXVIII of the series, by the properness route rather than any spectral one, confirming the warning of this paper.

1. WHAT THE ROUTE REQUIRES

We keep the notation of Paper X: $\mathcal{H}$ the Hurwitz order, $P = \mathcal{H} \setminus \{0\}$, $G = \mathbb{H}(\mathbb{Q})^\times$, Nrd the reduced norm, $\mathcal{T}(P)$ the Nica–Toeplitz algebra with isometries v_a , projections $e_a = v_a v_a^*$, dynamics $\sigma_t(v_a) = \text{Nrd}(a)^{it} v_a$, and φ an extremal KMS_β state with $\varphi(e_a) = \text{Nrd}(a)^{-\beta}$.

For an odd prime p let $a_1, \dots, a_{p+1}$ represent the $p+1$ principal right ideals of reduced norm p , which are the $p+1$ neighbours of a vertex in the $(p+1)$ -regular Bruhat–Tits tree of $SL_2(\mathbb{Q}_p)$, $\mathbb{H} \otimes \mathbb{Q}_p$ being split for p odd (Paper X).

Sources, code and data for the entire series: <https://github.com/Ruqing1963/localized-bost-connes>; this paper is in [papers/XI-C4/](#).

The series. The foundational paper is [8]; Papers II, III, V, IX and X, cited by numeral, are in preparation, and the facts we use from them are re-proved where they are used.

The route under examination is: (i) form $\mathcal{M}_p = \frac{1}{p+1} \sum_k v_{a_k}^*(\cdot) v_{a_k}$; (ii) identify it with a tree adjacency operator; (iii) invoke Deligne's bound to get $\|\mathcal{M}_p\|_{L_0^2} \leq 2\sqrt{p}/(p+1) < 1$; (iv) conclude non-amenability by a Kesten-type criterion. We show that (ii) and (iii) both fail, and that repairing (ii) destroys the system.

2. THE PROJECTIONS DO NOT PARTITION

Proposition 2.1 (Nested, not orthogonal). *Let $j \neq k$. Then*

$$e_{a_j} e_{a_k} = e_{a_j \vee a_k} \neq 0, \quad \varphi(e_{a_j} e_{a_k}) = p^{-2\beta} > 0, \quad \sum_{k=1}^{p+1} \varphi(e_{a_k}) = (p+1)p^{-\beta}.$$

In particular $\{e_{a_k}\}$ is not an orthogonal family and $\sum_k e_{a_k}$ is not a scalar, so it is not a partition of unity for any β .

Proof. In the Nica-Toeplitz algebra $e_a e_b$ is the projection onto $aP \cap bP$, which is $(a \vee b)P$ by the LCM property, nonempty by the Ore property (Paper X). For distinct right ideals of reduced norm p one has $a_j \mathcal{H} \cap a_k \mathcal{H} = p\mathcal{H}$: the central element $p = a_j \bar{a}_j = a_k \bar{a}_k$ lies in both, the two ideals are distinct maximal right ideals containing $p\mathcal{H}$, so $a_j \mathcal{H} + a_k \mathcal{H} = \mathcal{H}$ and $[\mathcal{H} : a_j \mathcal{H} \cap a_k \mathcal{H}] = [\mathcal{H} : a_j \mathcal{H}][\mathcal{H} : a_k \mathcal{H}] = p^4 = [\mathcal{H} : p\mathcal{H}]$. Hence $a_j \vee a_k = p$ up to units, $e_{a_j} e_{a_k} = e_p$ and $\varphi(e_p) = \text{Nrd}(p)^{-\beta} = p^{-2\beta}$. The last identity is $\varphi(e_{a_k}) = p^{-\beta}$ summed over $p+1$ terms. $\square$

Corollary 2.2 (The tree picture is unavailable). *The adjacency operator of a $(p+1)$ -regular tree acts on ℓ^2 of the vertex set by summing over $p+1$ disjoint branches, and its analysis — Kesten's $2\sqrt{p}$, Deligne's bound for the LPS quotients — rests on that disjointness. By Proposition 2.1 the branches of $\mathcal{T}(P)$ overlap, with overlap mass $p^{-2\beta}$ at every pair. No identification of $\mathcal{M}_p$ with a tree adjacency operator is therefore available, and the Ramanujan bound has no object to bound.*

Remark 2.3 (Numerical illustration). At $p = 3$: $\sum_k \varphi(e_{a_k}) = 1.3333$ and $\varphi(e_j e_k) = 0.1111$ for $\beta = 1$; 0.4444 and 0.01235 for $\beta = 2$. At $p = 13$, $\beta = 2$: 0.0828 and $3.5 \cdot 10^{-5}$. The overlaps shrink as β grows but never vanish, and the branch masses never sum to 1.

Remark 2.4 (This is exactly the property we wanted). The failure is not incidental. The Ore and LCM conditions were imposed in Paper X precisely to guarantee that no Cuntz relation holds, which is what keeps the high-temperature region $\beta < 2$ from being excluded, as it is in the affine systems of Paper II and the free variant of Paper V. Nestedness is the feature that makes the Hurwitz model viable; and nestedness is the feature that forbids the tree. The two requirements are in direct opposition.

3. THE MARKOV OPERATOR DOES NOT CONTRACT

Proposition 3.1 ($\mathcal{M}_p$ is unital but not state-preserving). $\mathcal{M}_p(x) = \frac{1}{p+1} \sum_k v_{a_k}^* x v_{a_k}$ is unital and completely positive, and

$$(3.1) \quad \varphi(\mathcal{M}_p(x)) = \frac{p^\beta}{p+1} \varphi\left(x \sum_k e_{a_k}\right) \quad (x \in \mathcal{T}(P)).$$

Since $\sum_k e_{a_k}$ is not a scalar by Proposition 2.1, $\varphi \circ \mathcal{M}_p \neq \varphi$. Consequently $\mathcal{M}_p$ induces on $L^2(\mathcal{T}(P), \varphi)$ only the bound

$$\|\mathcal{M}_p\|_{L^2(\varphi)} \leq p^{\beta/2},$$

which exceeds 1 for every $\beta > 0$, and no contraction — still less a gap below $2\sqrt{p}/(p+1)$ — follows from these relations.

Proof. Unitality is $\frac{1}{p+1} \sum_k v_{a_k}^* v_{a_k} = 1$; complete positivity is clear. The KMS condition with the entire element v_a^* , $\sigma_{i\beta}(v_a^*) = \text{Nrd}(a)^\beta v_a^*$, gives $\varphi(v_a^* x v_a) = \text{Nrd}(a)^\beta \varphi(x e_a)$, the scaling identity of Paper III, whence (3.1) with $\text{Nrd}(a_k) = p$. For the bound: e_a is fixed by σ , so $\varphi(e_a y) = \varphi(y e_a)$ for all y , whence $\varphi(x e_a) = \varphi(e_a x e_a)$ and, for $x \geq 0$, $\varphi(x) = \varphi(e_a x e_a) + \varphi(e_a^\perp x e_a^\perp) \geq \varphi(e_a x e_a)$; so $\varphi \circ \mathcal{M}_p \leq p^\beta \varphi$ on positive elements, and a unital completely positive map Φ with $\varphi \circ \Phi \leq C\varphi$ satisfies $\varphi(\Phi(x)^* \Phi(x)) \leq \varphi(\Phi(x^* x)) \leq C\varphi(x^* x)$ by the Kadison–Schwarz inequality, i.e. extends to $L^2(\varphi)$ with norm at most $C^{1/2}$. $\square$

Remark 3.2. Proposition 3.1 is the Hurwitz instance of the phenomenon recorded in Paper III (its Proposition 5.1): the transfer operator $x \mapsto v_a^* x v_a$ never preserves a KMS_β state, because the KMS condition converts it into $\text{Nrd}(a)^\beta \varphi(\cdot e_a)$. Whatever spectral theory one wishes to do with $\mathcal{M}_p$ must first address this, and normalizing by $(p+1)^{-1}$ does not.

4. THE SPECTRAL GAP STATEMENT IS NOT WELL POSED

Proposition 4.1 (C1 is not invariant). *For the non-singular action of G on (Ω, ν_β) with $\nu_\beta(a \cdot) = \text{Nrd}(a)^{-\beta} \nu_\beta$, the Koopman representation is*

$$(\pi(a)\xi)(x) = \text{Nrd}(a)^{\beta/2} \xi(a^{-1}x), \quad \text{so} \quad \pi(a)1 = \text{Nrd}(a)^{\beta/2} \mathbf{1}_{a\Omega} \neq 1$$

for $a \notin P^\times$. Hence $\mathbb{C}1$ is not a G -invariant subspace, $L_0^2 = L^2 \ominus \mathbb{C}1$ is not a subrepresentation, and the quantity $\|\mathcal{M}_p|_{L_0^2}\|$ is not the norm of an operator on an invariant subspace.

Proof. Immediate from the Radon–Nikodym cocycle; $\mathbf{1}_{a\Omega} \neq 1$ because $a\Omega$ is a proper clopen subset, of measure $\text{Nrd}(a)^{-\beta} < 1$. $\square$

Remark 4.2 (Why this matters). For a *measure-preserving* action the decomposition $L^2 = \mathbb{C}1 \oplus L_0^2$ is invariant and a spectral gap on L_0^2 is meaningful and does obstruct amenability. That is the setting of Kesten’s criterion and of the standard argument “non-amenable group acting measure-preservingly $\Rightarrow$ non-amenable action”. Neither applies here: our measure is quasi-invariant with a nontrivial cocycle, which is the whole point of a KMS_β state, and it is precisely for non-singular actions that non-amenable groups routinely admit amenable actions [1, 7] — the reason (C4) was posed as open in Paper X rather than settled there.

5. FORCING THE PARTITION DESTROYS THE SYSTEM

Theorem 5.1 (Forcing the partition kills the algebra). *Let A be a unital C^* -algebra generated by isometries v_a , $a \in P$, with $v_a v_b = v_{ab}$, and let p be an odd prime. If two range projections e_{a_j}, e_{a_k} , $j \neq k$, of reduced norm p are orthogonal in A , then $A = 0$. In particular the “boundary quotient” $\partial\mathcal{T}(P)$ of $\mathcal{T}(P)$ by the relations $\sum_{k=1}^{p+1} e_{a_k} = 1$, for any single odd prime p , is the zero algebra, and carries no state, KMS or otherwise.*

Proof. Since $\text{Nrd}(a_j) = p$ we have $p = a_j \bar{a}_j$ with $\bar{a}_j \in P$, so $v_p = v_{a_j} v_{\bar{a}_j}$ and $e_p = v_p v_p^* = v_{a_j} e_{\bar{a}_j} v_{a_j}^* \leq v_{a_j} v_{a_j}^* = e_{a_j}$; likewise $e_p \leq e_{a_k}$. If $e_{a_j} e_{a_k} = 0$ then $e_p \leq e_{a_j} \wedge e_{a_k} = 0$, so $v_p = e_p v_p = 0$ and $1 = v_p^* v_p = 0$. A sum of projections equal to 1 is an orthogonal sum: $e_j = e_j \sum_k e_k$ gives $\sum_{k \neq j} e_j e_k e_j = 0$ with positive summands, so $e_j e_k e_j = 0$ and $e_j e_k = 0$. Hence the relation $\sum_k e_{a_k} = 1$ already implies $1 = 0$. $\square$

Remark 5.2 (Partition in mean). One might weaken the partition to a partition *in mean*, asking only that $\sum_k \varphi(e_{a_k}) = 1$ for a KMS_β state of $\mathcal{T}(P)$ itself. This forces $(p+1)p^{-\beta} = 1$, i.e.

$$\beta = \beta(p) := \frac{\log(p+1)}{\log p},$$

and $p \mapsto \beta(p)$ is strictly decreasing on $(1, \infty)$, since $\frac{d}{dp} \log(\log(p+1)/\log p)$ is negative there: $\beta(3) = 1.261860$, $\beta(5) = 1.113283$, $\beta(7) = 1.068622$, $\beta(11) = 1.036287$, $\beta(13) = 1.028893$, $\beta(101) = 1.002135$, $\beta(1009) = 1.000143$, decreasing to 1. Any two primes already give a contradiction, so not even the mean version of the partition is compatible with a KMS_β state at more than one prime.

Corollary 5.3 (The dichotomy). *For the Hurwitz system the following are mutually exclusive:*

- (1) *the range projections at some prime partition the unit, so that tree and Ramanujan methods apply;*
- (2) *the algebra is nonzero, in particular a KMS_β state exists.*

Consequently no argument that first passes to a partition can decide (C4), the object it would decide being zero.

Remark 5.4 (Contrast with the classical $ax+b$ case). In the affine systems of Paper II the Cuntz relation at $\mathfrak{p}$ reads $\text{Np} \cdot \text{Np}^{-\beta} = 1$, giving $\beta = 1$ *independently of $\mathfrak{p}$* , so a KMS state survives at the single temperature $\beta = 1$; there the Np translates of a fundamental domain are genuinely disjoint, the acting semigroup being $ax+b$ rather than multiplicative. Here the multiplicity is $p+1$ while the norm is p , so the forced temperature $\log(p+1)/\log p$ depends on p and no temperature survives. The mismatch between multiplicity and norm — $p+1$ against p , the discrepancy being exactly the ramification-free part of the Bruhat–Tits degree — is what makes the Hurwitz boundary empty where the affine boundary is not.

6. THE CORRECT REFORMULATION

Proposition 6.1 (Maharam). *(C4) is equivalent to non-amenability of the Maharam extension [5]*

$$G \curvearrowright (\tilde{\Omega} \times \mathbb{R}, \tilde{\nu}_\beta \otimes e^s ds), \quad a \cdot (x, s) = (ax, s + \beta \log \text{Nrd}(a)),$$

on the dilation $\tilde{\Omega}$ carrying the G -action and the extended measure $\tilde{\nu}_\beta$, which preserves the displayed infinite measure.

Proof. Invariance of the displayed measure: the pushforward under a scales $\tilde{\nu}_\beta$ by $\text{Nrd}(a)^{-\beta}$ and $e^s ds$ by $e^{\beta \log \text{Nrd}(a)}$, which is the construction of the Radon–Nikodym cocycle. For the equivalence: amenability of a measured groupoid is equivalent to injectivity of its von Neumann algebra [1], and the von Neumann algebra of the Maharam extension is the continuous core $M \rtimes_{\sigma^\varphi} \mathbb{R}$ of $M = \pi_\varphi(\mathcal{T}(P))''$ [6, Ch. XII]. If M is injective so is $M \rtimes_\sigma \mathbb{R}$, a crossed product by an amenable group; if $M \rtimes_\sigma \mathbb{R}$ is injective then so is $M \otimes B(L^2(\mathbb{R})) \cong (M \rtimes_\sigma \mathbb{R}) \rtimes_{\hat{\sigma}} \mathbb{R}$ by Takesaki duality, hence M . So M is injective iff the Maharam extension is amenable. $\square$

Remark 6.2 (Why this is progress, and why it is not a solution). The reformulation removes the obstruction of §4: on the Maharam extension the measure is invariant, so decompositions of L^2 into invariant pieces and spectral gap arguments are meaningful. It does not resolve (C4). Amenability of an infinite-measure-preserving action of a non-amenable group is possible — the translation action of G on itself with counting measure is amenable as a groupoid, being principal and proper — so non-amenability of $G = \mathbb{H}(\mathbb{Q})^\times$ still does not transfer. What would be needed is a spectral gap for the Koopman representation of the Maharam extension on the orthogonal complement of the invariant functions, and we have not computed it.

7. ASSESSMENT

Step of the proposed route	Status	Reference
$p + 1$ ideals $\leftrightarrow$ tree neighbours	true	Paper X
$\mathbb{H}(\mathbb{Q})^\times$ non-amenable	true, necessary	Paper X
the e_{a_k} partition	false: nested	Prop. 2.1
$\mathcal{M}_p$ a tree adjacency operator	false	Cor. 2.2
$\mathcal{M}_p$ preserves φ	false ; $\ \mathcal{M}_p\ \leq p^{\beta/2}$	Prop. 3.1
L_0^2 a subrepresentation	false	Prop. 4.1
$\ \mathcal{M}_p _{L_0^2}\ \leq 2\sqrt{p}/(p+1)$	not well posed	Prop. 4.1
pass to the boundary to force a partition	the quotient is zero	Thm. 5.1
(C4) resolved	no	—
(C4) $\iff$ Maharam non-amenable	true reformulation	Prop. 6.1

The route is closed, and Corollary 5.3 says something stronger than that this particular attempt fails: *no* method requiring a partition of unity into $p+1$ pieces can decide (C4), because the object on which such a partition exists is zero. This rules out tree adjacency operators, random walks on the Bruhat–Tits building, and Ramanujan or Kesten bounds in the form used above. The feature that makes them applicable is exactly the Cuntz relation that the Ore and LCM conditions of Paper X were designed to exclude.

We are left with Proposition 6.1 as the reformulation to pursue and with the following as the concrete open questions. Does the Koopman representation of the Maharam extension on the complement of the invariant functions have a spectral gap? If it does not, (C4) fails and by Connes’ classification, as in Paper IX, the Hurwitz factor collapses to R_∞ , R_λ or a Krieger factor like all the others, which would be a definitive negative answer to the whole programme. If it does, one obtains the first non-injective arithmetic Bost–Connes factor, and the deformation/rigidity machinery that Paper IX found inapplicable for want of a spectral gap becomes actionable. Second: is there a variant of the Hurwitz system in which the multiplicity $p+1$ and the norm p are brought into agreement, so that a boundary quotient with a nonempty KMS simplex exists? Remark 5.4 suggests looking for an order whose ideals of norm $N\mathfrak{p}$ number exactly $N\mathfrak{p}$; whether such an order exists we do not know, and it is the one place where the dichotomy of Corollary 5.3 might be broken rather than circumvented.

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